

TandemBench: A Joint-Corruption Benchmark & Analysis of Cross-Modal Retrieval

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Abstract

Vision-language retrieval systems may encounter degradation in both images and text simultaneously, yet robustness is commonly evaluated one modality at a time. This leaves a basic question unresolved: can joint degradation be inferred from the corresponding image-only and text-only effects? We introduce TandemBench, a benchmark for matched marginal and joint corruption that evaluates 19 dual-encoder retrieval models across 16 image perturbations, 12 text perturbations, five ordinal severity levels, two datasets, and both retrieval directions. We separate perturbation applicability, execution, and marginal difficulty, and use an eight-operator execution-screened subset containing 47,360 complete conditions for primary analysis. Joint degradation exceeds the larger single-modality loss in 95.15% of these conditions, by 5.07 Recall@1 points on average; the same qualitative result holds across the full twelve-operator suite. More importantly, the sum of the two marginal losses reconstructs joint degradation closely under mild corruption, but its predictive accuracy declines as degradation strengthens and its error becomes dependent on retrieval cutoff and model. Realized-damage matching shows that these conclusions are not an artifact of pairing operators by a common severity index. Recovery experiments further reveal strong dependence on retrieval direction, corruption, and model. We release TandemBench, its evaluation harness, corruption implementations, configurations, and all 83,754 evaluation records for reproducible joint-corruption analysis.

1. Introduction

Wearable cameras, on-device assistants, and other multimodal systems increasingly rely on image-text retrieval to connect everyday visual scenes with natural-language queries. Datasets such as MS-COCO and Flickr30K capture many of these ordinary objects, activities, and environments, but deployment introduces an additional challenge:

the visual stream may be degraded by motion blur, low light, glare, haze, or compression, while the accompanying text may contain misspellings, dropped words, transposed characters, or incorrect terms. Retrieval may also feed downstream search, summarization, reasoning, or decision support, allowing upstream errors to propagate through the system. Because these degradations can occur simultaneously, retrieval models must remain reliable when neither modality is clean.

Existing robustness work studies image and text degradation in vision-language systems [14, 16, 18], including recent work on multimodal corruption and test-time robustness [1, 3, 10, 25]. The missing question for bidirectional retrieval is narrower: *if image-only and text-only robustness have already been measured, how much do those marginal measurements tell us about performance when both modalities degrade together?* Answering it requires matched image-only, text-only, and joint conditions on the same samples and clean baseline. This separates two questions: whether either marginal alone understates joint degradation, and whether the *pair* of marginals can reconstruct it.

We introduce **TandemBench**, which evaluates matched triples of image-side degradation δ_1 , text-side degradation δ_2 , and joint degradation δ_3 across 19 dual-encoder models, MS-COCO and Flickr30K, 16 image corruptions, 12 text perturbations, five ordinal severity levels, and both retrieval directions. Because a requested text perturbation may not actually modify a short caption, we separately characterize applicability, execution, and marginal difficulty; all twelve operators remain in the benchmark, while an execution-screened subset provides the primary analysis.

Across 47,360 primary conditions, joint degradation exceeds the larger marginal in 95.15% of cases by 5.07 Recall@1 points on average; the conclusion also holds over the full twelve-operator suite (91.84%, +3.79 points). More importantly, $\delta_1 + \delta_2$ reconstructs joint degradation closely under mild corruption, but condition-level predictability declines as degradation strengthens and the direction of error becomes retrieval-cutoff- and model-dependent. The finding persists when conditions are matched by realized

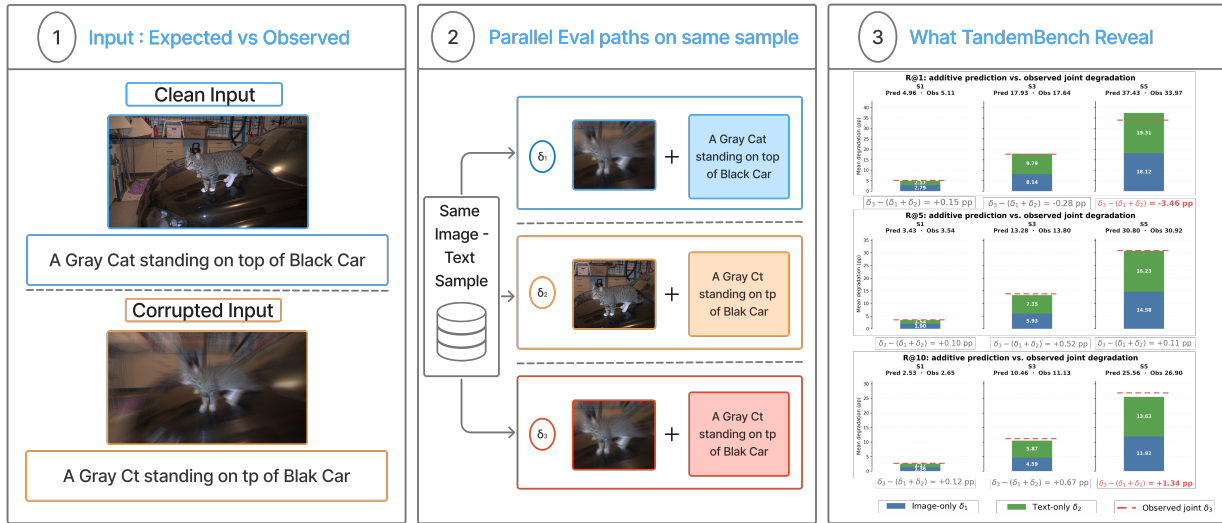


Figure 1. **Overview of TandemBench.** The same image–text sample is evaluated along image-only, text-only, and joint-corruption paths. On the eight-operator execution-screened subset, joint degradation exceeds the larger marginal loss in 95.15% of 47,360 complete conditions, with a mean excess of 5.07 Recall@1 points. TandemBench additionally tests whether the two marginal losses can reconstruct the joint effect.

marginal damage rather than a common severity index. Recovery experiments further show that methods transferring under conventional shift can become direction- and model-dependent when both retrieval sides are degraded.

Our contributions are:

- **Matched marginal–joint benchmark:** 19 models, two datasets, 16 image corruptions, 12 text perturbations, five severities, and both retrieval directions.
- **Execution-aware perturbations:** applicability, fire rate, and marginal difficulty are measured separately rather than conflated with validity.
- **Marginal-to-joint composition:** we quantify both degradation missed by either marginal and when the two marginals reconstruct the joint effect.
- **Recovery and release:** we stress-test recovery under joint corruption and release the harness, configurations, corruptions, protocol, and all 83,754 evaluation records.

2. Related Work

Natural-corruption robustness was formalized by ImageNet-C [6], while text robustness developed through character-level noise, semantic substitution, behavioural testing, and perturbation toolkits [2, 8, 13, 15, 21]. These lines established many corruption families used in TandemBench, but their basic evaluation unit is a single degraded channel.

Vision–language robustness has since expanded to image- and text-side retrieval perturbations [14, 18],

broader VLM corruption benchmarks [10, 16], simultaneous multimodal test-time shift [3], and joint adversarial attacks [12, 23]. These studies show that both modalities matter, but TandemBench asks a different question: every joint condition is paired with its image-only and text-only counterparts on the same samples and clean baseline, yielding matched $(\delta_1, \delta_2, \delta_3)$ measurements for testing whether either marginal alone captures joint loss and whether the pair reconstructs it. A detailed comparison with prior benchmarks is provided in the supplement.

A related methodological issue is whether a requested perturbation is actually exercised by the data. Compositionality benchmarks such as ARO, SugarCrepe, and Winoground construct examples around the linguistic structures they test [7, 19, 22], whereas generic corruption operators may be inapplicable to short captions. TandemBench therefore separates applicability, fire rate, and marginal difficulty.

Test-time adaptation is complementary. Established approaches include entropy minimization, augmentation-based adaptation, prompt tuning, and cache- or feature-based procedures [5, 9, 17, 20, 24]; retrieval-specific methods such as TCR [11] and UCA [25] target cross-modal shift directly. These methods motivate our recovery experiments, which we treat as a downstream stress test rather than the benchmark’s defining contribution.



Figure 2. **Example perturbation schedules.** Motion blur and keyboard-typo corruption across five severity levels.

3. The Benchmark

TandemBench isolates image-side, text-side, and simultaneous degradation while keeping the retrieval task fixed. Models are evaluated without parameter updates in the main benchmark, and corrupted sets are generated deterministically from public source data.

3.1. Source Data

TandemBench uses MS-COCO and Flickr30K, two established image-text retrieval benchmarks. We evaluate the 5,000-image COCO val2017 split with 25,014 captions and the 1,000-image Karpathy test split of Flickr30K with 5,000 captions.

Both datasets provide multiple captions per image. COCO retrieval is often reported on the Karpathy 5K test split rather than val2017; our corruption effects remain internally comparable because every result is measured relative to the clean baseline on the same split, although absolute clean scores should not be compared directly across split conventions.

3.2. Perturbation Suites

Image corruption. We use sixteen operators at five severity levels following ImageNet-C [6]: noise (Gaussian, shot, impulse, speckle), blur (defocus, glass, motion, zoom), weather (snow, frost, fog, brightness), and digital corruption (contrast, elastic transform, pixelate, JPEG compression). Fifteen are standard ImageNet-C corruptions and speckle noise comes from the extended set. These operators are defined for every image and therefore require no structural eligibility check; all sixteen are retained throughout.

Text corruption. We use twelve perturbations drawn from common NLP robustness taxonomies [13, 15, 21]: character-level (keyboard typo, OCR error, character swap, character delete), word-level (word delete, word shuffle, punctuation noise), and semantic (negation insertion, attribute swap, relation swap, object substitution, sentence shuffle).

Text perturbations differ in whether the required linguistic structure is present and whether the requested transformation is actually executed. We therefore characterize each

operator using three quantities: *applicability*, whether the caption contains the structure required by the operator; *fire rate*, the fraction of captions that are actually modified; and *marginal difficulty*, the resulting text-side retrieval degradation. These quantities are reported separately rather than treating model degradation as a criterion for perturbation validity.

All twelve operators constitute the full TandemBench text suite. For analyses requiring perturbations that execute broadly on short retrieval captions, we additionally define an *execution-screened subset* containing the eight operators whose severity-averaged fire rate exceeds 55%. For analyses specifically intended to stress non-trivial marginal degradation, we report a nested seven-operator *high-damage challenge subset*, which additionally requires more than 8 Recall@1 points of text-side degradation. The damage criterion defines only this challenge subset; it does not determine whether an operator belongs to TandemBench. Sensitivity to these analysis subsets is evaluated in Sec. 4.2 and the supplement.

Severity is operator-specific. Character-level operators modify 1, 3, 5, 7, and 10% of characters across severity levels 1–5; word deletion and punctuation noise affect 5–40% of tokens; word shuffle affects 10–60%; and semantic operators use schedules spanning 10–80% of eligible tokens or sentences. These levels order intervention strength within each operator; their indices do not imply equal physical or linguistic distortion across different operators.

3.3. Models

The benchmark includes nineteen dual-encoder retrieval models from eight families, spanning variation in scale, pretraining data, and training objective. This diversity allows us to test whether the degradation missed by single-modality evaluation is specific to particular models or persists across substantially different retrieval systems. Five families include multiple scales, enabling within-family analysis of how this gap changes with model capacity and compute.

CLIP, MetaCLIP, EVA-CLIP, and SigLIP keep the pretraining data source fixed across scales. OpenCLIP is the exception: its B/16 checkpoint uses LAION-400M, whereas the remaining OpenCLIP variants use LAION-2B checkpoints. Clean Recall@1 spans 49.96–79.64 on COCO image-to-text retrieval and 78.80–94.20 on Flickr30K. Full clean baselines are provided in the supplement.

All evaluated systems are dual encoders; no multimodal LLMs are included. BLIP-large and CoCa ViT-L are fine-tuned for COCO retrieval, and BLIP is evaluated without image-text matching reranking so that all models use the same dual-encoder similarity protocol.

3.4. Evaluation Protocol

Every corruption condition is evaluated along three matched paths. Path 1 corrupts the image only, producing image-side degradation δ_1 ; Path 2 corrupts the caption only, producing text-side degradation δ_2 ; and Path 3 corrupts both modalities on the same sample, producing joint degradation δ_3 . Each quantity is measured as the drop from the same clean baseline for that model, dataset, retrieval direction, and retrieval metric, making the three effects directly comparable within a condition.

Image and text corruptions are paired by a common ordinal severity index, $s \in \{1, \dots, 5\}$. Thus, an image corruption at severity s is evaluated jointly with a text corruption at the same severity index. This traces the diagonal of the full image–text severity grid and provides a consistent protocol for increasing both interventions together. The shared index does not imply that two different operators at the same severity produce equal physical, linguistic, or realized performance degradation. We therefore additionally test whether the main conclusions persist when conditions are matched by their realized marginal damage rather than by severity index.

The basic analysis unit is a complete triple $(\delta_1, \delta_2, \delta_3)$ indexed by model, dataset, retrieval direction, image operator, text operator, and severity. Unless otherwise stated, non-normalized quantities, including marginal degradation, joint degradation, excess over the larger marginal, and additive prediction error, are computed over all available complete triples. A minimum-damage condition of $\delta_1 + \delta_2 \geq 0.5$ percentage points is applied only to normalized ratio analyses, where near-zero marginal damage can make the denominator unstable.

The release contains 83,754 evaluation records in total. The full twelve-operator text suite yields 71,040 complete matched conditions. The eight-operator execution-screened subset defined in Sec. 3.2 yields 47,360 complete conditions and serves as the primary analysis subset. The nested seven-operator high-damage challenge subset contains 41,440 complete conditions; applying the 0.5-pp minimum-damage rule for normalized analyses reduces this subset to 41,264 conditions. Raw and additive analyses do not use this reduction.

Both retrieval directions are evaluated independently. In image-to-text retrieval, image corruption affects the query and text corruption affects the gallery; in text-to-image retrieval the assignment reverses. Under joint corruption, both query and gallery are degraded.

3.5. Metrics and Reproducibility

Recall@1 is our primary retrieval metric. We additionally evaluate Recall@5 and Recall@10 to test whether the principal conclusions depend on retrieval cutoff. For each metric, δ_1 , δ_2 , and δ_3 denote the corresponding drops from the

same clean baseline.

We compare joint degradation with two distinct references. The excess

$$\delta_3 - \max(\delta_1, \delta_2)$$

measures how much degradation would be missed by considering only the worse single-modality condition. The additive prediction error

$$\delta_3 - (\delta_1 + \delta_2)$$

instead measures whether the two marginal losses are sufficient to reconstruct the joint effect. These quantities answer different questions: the first evaluates the incompleteness of either marginal alone, while the second evaluates how the two marginal effects compose. Relative summaries such as $\delta_3 / \max(\delta_1, \delta_2)$ are treated as descriptive and use the minimum-damage safeguard defined in Sec. 3.4.

Confidence intervals for condition-level summaries use 50,000 clustered bootstrap resamples over model–image–corruption–text–perturbation groups, preserving repeated measurements across severity, dataset, and retrieval direction within each cluster. These intervals quantify uncertainty over the observed benchmark structure rather than generalization to arbitrary unseen models or corruption operators. As a more conservative sensitivity analysis, we additionally resample the model, image–corruption, and text–perturbation axes separately. For the primary Recall@1 result, this widens the mean-excess interval from [4.96, 5.17] to [3.64, 6.41] points and the exceedance interval from [94.78, 95.52]% to [91.46, 98.15]%; both remain positive and preserve the qualitative conclusion. We report these crossed-axis intervals as sensitivity rather than as a guarantee for arbitrary unseen models or operators. Rank comparisons use Spearman correlation with bootstrap resampling over models, while severity trends use permutation tests.

All corruptions are generated deterministically. We release the evaluation code, corruption configurations, model configurations, and complete evaluation records required to regenerate the reported analyses.

4. Perturbation Screening and Joint-Corruption Analysis

4.1. Do the Perturbations Perturb?

Before comparing marginal and joint degradation, we ask whether a requested text perturbation is actually exercised by the data. We measure **fire rate**, the fraction of 500 captions per operator and severity that are modified, and for structure-dependent operators **applicability**, the fraction containing the required linguistic structure. Marginal retrieval degradation is reported separately as difficulty.

Table 1 separates execution from difficulty. punctuation.noise, for example, fires on 56.0%

Table 1. **Text perturbation characterization.** Fire rate is the fraction of captions modified by an operator; Eligible is the fraction containing the required linguistic structure. The execution-screened subset uses mean fire rate $> 55\%$. The nested challenge subset additionally requires $\delta_2 > 8$ pp. Marginal damage is reported as difficulty, not as a validity criterion.

Operator	Class	Fire %	S5 %	Elig. %	δ_2 pp	Exec.	Chal.
word shuffle	surface	99.6	100	100.0*	8.82	✓	✓
word delete	surface	75.0	99	100.0*	11.69	✓	✓
char delete	surface	74.4	98	100.0*	11.68	✓	✓
char swap	surface	74.0	98	100.0*	14.55	✓	✓
object substitution	semantic	70.6	98	100.0	12.57	✓	✓
keyboard typo	surface	69.8	97	100.0*	12.01	✓	✓
ocr error	surface	59.8	90	100.0*	8.67	✓	✓
punctuation noise	surface	56.0	90	100.0*	1.25	✓	–
relation swap	semantic	25.6	41	53.4	0.32	–	–
attribute swap	semantic	18.2	30	50.9	3.89	–	–
negation insertion	semantic	4.0	9	41.3	0.65	–	–
sentence shuffle	semantic	0.0	0	5.2	0.26	–	–

* Applicability is 100% by construction. Exec.: execution-screened subset; Chal.: high-damage challenge subset.

of captions (90% at severity 5) yet induces only 1.25 Recall@1 points of text-side degradation, so low damage is a robustness result rather than invalidity. By contrast, several semantic operators have low applicability because short captions often lack the relations, attributes, negation contexts, or multi-clause structure they require. All twelve remain in the full suite; eight pass the execution screen and seven additionally form the high-damage challenge subset.

4.2. How Operator Composition Changes Conclusions

Operator selection affects both normalized interaction measures and aggregate conclusions. We first tested three candidate normalized interaction estimators against text-side marginal degradation under both per-condition and aggregate-first computation. Sum-normalized and difference-based forms remain strongly associated with perturbation strength, with correlations reaching $\rho = -0.951$; their largest values occur for operators whose text-side damage is near zero. Normalizing by the larger marginal is less sensitive but still changes with aggregation order. Near-zero marginal damage can therefore make normalized interaction scores reflect denominator scale as much as composition.

We consequently keep δ_1 , δ_2 , and δ_3 explicit and use the unnormalized additive error $\delta_3 - (\delta_1 + \delta_2)$ when asking whether the two marginals reconstruct the joint condition. Normalized ratios remain descriptive and use the 0.5-pp safeguard from Sec. 3.4; full estimator diagnostics and aggregation-order sensitivity are reported in the supplement.

Importantly, the principal joint-versus-marginal result does not depend on the high-damage challenge subset.

Using all twelve text operators and all complete triples, joint degradation exceeds the larger marginal in 91.84% of 71,040 conditions, with a mean excess of +3.79 Recall@1 points. Restricting only by execution rate yields 47,360 conditions across eight text operators and increases these values to 95.15% and +5.07 points. The nested seven-operator high-damage challenge subset contains 41,440 complete conditions and yields 96.15% and +5.63 points. Thus, emphasizing stronger perturbations increases the magnitude of the effect, but the qualitative conclusion is already present without damage-based operator selection.

Operator composition can still change modality-fragility summaries: 16 of 19 models appear image-dominant over all twelve operators versus only 4 of 19 on the high-damage subset, while model ordering remains much more stable ($\rho = 0.968$). We therefore report the full suite together with explicit primary and challenge subsets.

4.3. How Much Single-Modality Evaluation Misses

We first ask how much degradation would be missed if robustness were evaluated on only one modality at a time. For each condition, we compare joint degradation with the larger of the two corresponding marginals, $\max(\delta_1, \delta_2)$. This gives single-modality evaluation its stronger available reference and measures whether either marginal alone captures the degradation observed when both inputs deteriorate. It is not a test of interaction or superadditivity.

On the eight-operator execution-screened subset, joint degradation exceeds the larger marginal in **95.15%** of 47,360 complete conditions, with a mean excess of **5.07 Recall@1 points** (95% clustered-bootstrap CI [4.96, 5.17]). Thus, although either marginal can capture a substantial part of the loss, evaluating only the more damaging modality still understates the realized joint degradation in most conditions. Section 4.2 shows that the same qualitative conclusion holds for both the full twelve-operator suite and the nested high-damage challenge subset.

The degree of understatement increases with severity. Exceedance rises from 88.64% at severity 1 to 99.33% at severity 5, while mean excess increases from 1.34 to 9.18 Recall@1 points. Joint degradation therefore becomes progressively less well represented by either marginal considered in isolation as both interventions strengthen.

This result should not be interpreted as evidence that joint corruption is generally synergistic. Joint degradation exceeds the *sum* of the two marginals in 39.29% of conditions overall, and this fraction declines from 55.00% at severity 1 to 22.69% at severity 5. The larger-marginal comparison therefore quantifies degradation missed by single-modality evaluation; whether the *pair* of marginal measurements can reconstruct the joint effect is a separate question addressed in Sec. 4.5.

Table 2. **Joint degradation resolved by dataset and retrieval direction.** Exceedance is the fraction of conditions in the execution-screened subset for which $\delta_3 > \max(\delta_1, \delta_2)$. Values below each rate show mean excess in Recall@1 points.

	I2T	T2I
MS-COCO	96.49% +5.48 pp	96.04% +3.12 pp
Flickr30K	91.27% +6.44 pp	96.69% +5.31 pp

4.4. Does the Effect Hold Across Datasets and Retrieval Directions?

Across all four dataset-direction strata, exceedance remains 91.27–96.69% with positive mean excess of +3.12 to +6.44 Recall@1 points. The pooled result is therefore not driven by one dataset or retrieval direction; frequency and magnitude need not rank strata identically.

4.5. When Do Marginals Reconstruct Joint Degradation?

The larger-marginal comparison asks whether either single-modality evaluation is sufficient. A stronger question is whether the *pair* of marginal measurements is sufficient: can the joint degradation δ_3 be reconstructed from $\delta_1 + \delta_2$? We evaluate this using the additive prediction error

$$\delta_3 - (\delta_1 + \delta_2),$$

where zero indicates exact additive composition, a positive value indicates joint degradation beyond the sum, and a negative value indicates that the sum overestimates the observed joint loss.

Under mild corruption, the two marginals reconstruct the joint effect remarkably well. At severity 1, for example, the Recall@1 marginals sum to 4.96 points against an observed joint degradation of 5.11, giving a bias of only +0.15 points and $R^2 = 0.968$. At severity 2, the corresponding values are 10.63 and 10.66 points, reducing the bias to +0.03. Recall@5 and Recall@10 show similarly strong agreement at low severity.

As corruption strengthens, condition-level predictability declines across all three retrieval cutoffs, but the *direction* of the departure from additivity is metric-dependent. At severity 5, the Recall@1 additive estimate is 37.43 points compared with 33.97 observed, producing a -3.46 point bias and reducing R^2 to 0.759. Recall@5, however, remains nearly additive at the same severity: 30.80 points predicted versus 30.92 observed, for a bias of only +0.11 points and $R^2 = 0.895$. Recall@10 instead remains mildly above the additive estimate, with a +1.34 point bias and $R^2 = 0.918$.

The high-severity negative bias at Recall@1 is therefore not a universal property of joint corruption. Across the full execution-screened subset, mean additive error is -0.99

Table 3. **Additive composition across retrieval cutoffs.** Bias is $\delta_3 - (\delta_1 + \delta_2)$ in Recall points; R^2 measures condition-level agreement between the additive estimate and observed joint degradation. Results use the eight-operator execution-screened subset.

	Recall@1		Recall@5		Recall@10	
S	Bias	R^2	Bias	R^2	Bias	R^2
1	+0.15	0.968	+0.10	0.980	+0.12	0.980
2	+0.03	0.970	+0.25	0.980	+0.33	0.982
3	-0.28	0.948	+0.52	0.969	+0.67	0.973
4	-1.38	0.887	+0.52	0.950	+1.06	0.956
5	-3.46	0.759	+0.11	0.895	+1.34	0.918

points at Recall@1, compared with +0.30 at Recall@5 and +0.70 at Recall@10. The two marginals are consequently highly informative, particularly under mild degradation, but their composition cannot be assumed to follow a single rule across severity and retrieval cutoff. TandemBench identifies where simple additive reconstruction is sufficient and where direct joint evaluation remains necessary.

The same conclusion is not an artifact of pairing operators by a common severity index. Restricting the primary subset to conditions whose realized marginal damages differ by at most 2 Recall@1 points leaves 11,855 conditions; joint degradation still exceeds the larger marginal in 95.39% of them, with a mean excess of +4.44 points and a mean additive error of only -0.26 points. Thus, equal ordinal severity is a convenient benchmark protocol, not the source of the observed joint-corruption effect.

4.6. What Varies Across Models

The aggregate composition pattern masks substantial model-level variation. At Recall@1 severity 5, additive prediction error ranges from -7.04 points for openclip_vit_b32 to -0.58 for dfn_vit_h14. Within each family containing multiple variants, the high-severity error becomes less negative toward the higher-capacity or higher-compute configurations: $-6.95 \rightarrow -5.13 \rightarrow -2.45 \rightarrow -1.93$ for CLIP and $-7.04 \rightarrow -6.69 \rightarrow -3.79 \rightarrow -2.49 \rightarrow -1.18$ for OpenCLIP. MetaCLIP, EVA-CLIP, and SigLIP show the same directional pattern.

These within-family trends are descriptive rather than controlled scaling experiments. Variants differ in architecture and compute, and OpenCLIP B/16 uses a different pre-training corpus from the remaining OpenCLIP checkpoints. For example, ViT-B/32 and ViT-B/16 have nearly identical parameter counts but differ substantially in visual token count and compute. The relevant observation is therefore not that scale itself causes improved composition, but that models differ systematically in how closely their marginal

Table 4. **Models under joint corruption, with Recall@1 additive prediction error resolved by severity.** Values are means over both datasets and retrieval directions using the eight-operator execution-screened text subset. “Exceeds” is the percentage of conditions in which joint degradation exceeds the larger marginal. The final five columns report $\delta_3 - (\delta_1 + \delta_2)$.

Family	Model	M	Clean R@1	Img	Txt	Joint	Joint % clean	Exceeds %	Mean excess	Joint/ larger	S1	S2	S3	S4	S5
CLIP	clip_vit.b32	151	54.5	10.42	11.46	19.22	35.3	93.2	4.61	1.68	0.05	-0.83	-1.72	-3.87	-6.95
CLIP	clip_vit.b16	150	57.2	8.77	11.32	18.17	31.8	92.2	4.64	1.60	-0.09	-0.56	-1.24	-2.60	-5.13
CLIP	clip_vit.l14	428	60.3	6.64	11.35	17.05	28.3	85.5	4.35	1.50	-0.23	-0.40	-0.36	-1.27	-2.45
CLIP	clip_vit.l14_336	428	62.1	6.68	11.14	17.21	27.7	88.6	4.68	1.54	-0.06	-0.07	-0.12	-0.86	-1.93
OpenCLIP	openclip_vit.b32	151	61.6	13.48	11.20	22.18	36.0	96.8	5.57	1.65	-0.05	-0.41	-1.41	-3.56	-7.04
OpenCLIP	openclip_vit.b16	150	60.9	12.33	11.64	21.76	35.7	98.0	5.72	1.76	0.14	-0.22	-1.02	-3.27	-6.69
OpenCLIP	openclip_vit.l14	428	68.3	9.77	10.86	19.58	28.6	97.5	5.42	1.80	0.21	0.08	-0.32	-1.45	-3.79
OpenCLIP	openclip_vit.h14	986	70.7	8.12	10.20	17.88	25.3	96.3	5.19	1.75	0.23	0.31	0.28	-0.50	-2.49
OpenCLIP	openclip_vit.big14	2540	72.5	7.25	9.27	16.59	22.9	95.8	5.04	1.79	0.42	0.36	0.60	0.16	-1.18
MetaCLIP	metaclip_vit.b16	150	63.9	11.09	10.69	20.34	31.8	96.1	5.65	1.83	0.16	0.06	-0.58	-1.85	-4.99
MetaCLIP	metaclip_vit.l14	428	69.4	7.61	10.21	17.50	25.2	97.3	5.08	1.71	0.30	0.36	0.26	-0.62	-1.93
EVA-CLIP	evaclip_vit.b16	150	64.5	9.00	10.42	18.50	28.7	98.2	5.20	1.78	0.27	0.25	-0.20	-1.24	-3.68
EVA-CLIP	evaclip_vit.l14	428	69.4	6.02	9.94	15.85	22.8	96.8	4.48	1.60	0.44	0.35	0.34	-0.35	-1.31
SigLIP	siglip_vit.b16	203	69.0	12.99	9.38	21.09	30.5	95.8	5.64	1.62	0.27	0.17	-0.34	-1.77	-4.71
SigLIP	siglip_vit.l16	652	73.2	9.40	7.89	16.84	23.0	94.4	4.70	1.79	0.15	0.16	0.04	-0.59	-2.03
SigLIP	siglip_so400m	878	75.7	6.67	8.23	15.17	20.0	95.5	4.67	1.84	0.28	0.52	0.67	0.52	-0.66
CoCa	coca_vit.l	638	72.1	12.44	9.46	20.66	28.7	95.6	5.56	1.66	0.12	0.19	-0.41	-1.91	-4.18
DFN	dfn_vit.h14	986	74.3	7.38	8.21	15.81	21.3	97.3	4.73	1.93	0.31	0.35	0.61	0.43	-0.58
BLIP	blip_large	470	71.2	11.00	11.12	20.64	29.0	99.0	5.56	1.86	-0.19	-0.38	-0.66	-1.81	-4.39

losses reconstruct joint degradation.

The broader model-level pattern remains visible under the execution-screened subset. Joint degradation as a fraction of clean performance ranges from 36.0% to 20.0% across the roster and correlates negatively with parameter count (Spearman $\rho = -0.790$, $p < 0.0001$). The dispersion of additive prediction error across models also increases with Recall@1 severity: its standard deviation rises from 0.19 to 0.37, 0.68, 1.25, and 2.03 points across severities 1–5. Thus, models behave similarly under mild corruption, where additive reconstruction is broadly accurate, but diverge increasingly in how their marginal effects compose as corruption strengthens.

Single-modality rankings provide unequal proxies for the joint condition. Image-only degradation tracks the joint ranking strongly (Spearman $\rho = 0.912$, 95% bootstrap CI [0.715, 0.984]), whereas text-only degradation is less predictive ($\rho = 0.512$). The two marginal rankings themselves agree only weakly ($\rho = 0.221$, $p = 0.363$). Single-modality evaluations therefore contain useful information about joint robustness, particularly on the image side, but they do not fully determine either the joint ranking or the model-specific departure from additive composition.

5. Recovery Under Joint Corruption

Joint corruption differs from conventional single-modality shift because both sides of retrieval may be unreliable. We therefore use recovery as a downstream stress test rather than an adaptation leaderboard, evaluating MS-

COCO on the seven-operator high-damage challenge subset. **Tandem-Aug** averages image representations across augmented views without parameter updates. **Tandem-Align** is a parameter-free feature-space procedure that shifts and scales text features to match the image-feature mean and standard deviation before re-normalization. We compare these with QueryExpansion [4] and retrieval-specific TCR [11].

The results are strongly method- and direction-dependent. Tandem-Aug is the only procedure in Table 5 with positive pooled gains in both directions (+0.83 I2T, +1.65 T2I), whereas Tandem-Align substantially reduces accuracy. QueryExpansion shows the clearest reversal, improving I2T by +3.40 points but reducing T2I by 8.72; reversing direction changes which corrupted modality is expanded.

TCR is informative because its published setting corrupts the query while keeping the gallery clean. Across ten checkpoints, corrupted-query/clean-gallery I2T yields +5.90 points (95% CI [+5.07, +6.75]); clean-query/corrupted-gallery yields +1.83 [+1.40, +2.21], while joint corruption retains +4.73 [+4.36, +5.10]. The reverse direction is less stable: the pooled T2I effect is -0.03 [$-0.23, +0.17$], but all three evaluated OpenAI CLIP checkpoints improve (+1.11 to +1.79) whereas every evaluated checkpoint from another pretraining pipeline declines (-0.21 to -1.10). I2T gains remain positive for all ten checkpoints. Recovery conclusions therefore depend on both direction and model family. Absolute gains can also hide declining fractional recovery. TCR’s I2T gain rises from +3.17 points at severity 1 to

Table 5. **Recovery under joint corruption.** MS-COCO Recall@1 change relative to no adaptation; parentheses show 95% CIs. Coverage differs across methods.

Type	Method	I2T	T2I	N
study	Tandem-Aug	+0.83 [+0.74,+0.91]	+1.65 [+1.60,+1.70]	7/7
study	Tandem-Align	-13.28 [-13.67,-12.88]	-12.70 [-12.93,-12.46]	5/7
prior	QueryExpansion [4]	+3.40 [+3.24,+3.55]	-8.72 [-8.85,-8.60]	7/7
prior	TCR [11]	+4.73 [+4.36,+5.10]	-0.03 [-0.23,+0.17]	10

+6.22 at severity 5 while the fraction of corruption-induced loss recovered falls from 48.6% to 11.5%; Tandem-Aug similarly falls from 45.5% to 8.1%. Thus, a method can recover more Recall points at high severity while repairing a smaller fraction of the corruption-induced loss.

We complement model-side adaptation with direct input restoration. Across five models, twelve corruption pairs, and severities 1, 3, and 5, restoring the image recovers +0.25 Recall@1 points (95% CI [+0.09,+0.41]), restoring the caption +1.72 [+0.88,+2.62], and restoring both +2.00 [+1.12,+2.93]. Their interaction is +0.02 [-0.03,+0.07], providing no measurable gain beyond the sum of the individual restorations. Restoration is also corruption-specific: spelling correction recovers +6.36 points for keyboard typos but changes performance by -0.74 on object substitution and -0.44 on word deletion, where it does not invert the underlying corruption. Additional adaptation methods are reported in the supplement.

6. Conclusion

TandemBench asks when joint retrieval degradation can be inferred from measurements made on each modality separately. Across 47,360 primary conditions, joint degradation exceeds the larger marginal in 95.15% of cases by 5.07 Recall@1 points on average; across all twelve text operators the result remains 91.84% and +3.79 points. Yet the marginals remain informative: their sum closely reconstructs joint degradation under mild corruption, while predictability declines with severity and the direction of error becomes retrieval-cutoff- and model-dependent.

We therefore ship TandemBench with the following joint-corruption evaluation protocol:

1. **Check applicability and execution:** report whether required structure exists and whether the operator actually modifies the input.
2. **Separate execution from difficulty:** treat marginal degradation as benchmark difficulty, not a validity test; label high-damage analyses as challenge subsets.
3. **Match marginal and joint conditions:** measure image-only, text-only, and joint corruption on the same samples

and clean baseline; ordinal severity need not imply equal realized damage.

4. **Keep $\delta_1, \delta_2, \delta_3$ explicit:** use normalized ratios cautiously when marginal degradation is near zero.
5. **Use both joint references:** compare δ_3 with $\max(\delta_1, \delta_2)$ for missed degradation and with $\delta_1 + \delta_2$ for reconstruction.
6. **Resolve composition:** report behavior across severity, retrieval metric, dataset/direction, and models rather than only a pooled statistic.
7. **Test recovery jointly:** report direction, model coverage, absolute gain, and recovered fraction when both retrieval sides are degraded.

Our realized-damage analysis shows that the main result is not an artifact of common ordinal severity. The recovery study reinforces the protocol: Tandem-Aug improves both directions, QueryExpansion changes sign, Tandem-Align degrades both, and TCR is strong in I2T but unstable in T2I. Results obtained under a single query/gallery shift therefore cannot be assumed to transfer when both sides are degraded.

The scope is deliberately bounded. We evaluate the diagonal of a five-level image-text severity grid rather than every asymmetric severity pair; realized-damage matching tests sensitivity to this choice but is not a physical or perceptual calibration of corruption strength. COCO and Flickr30K contain relatively short English captions, so structure-dependent semantic operators are often applicable only to a subset of examples; eligibility-conditioned evaluation and human semantic audits are natural extensions. All 19 systems are dual encoders, leaving cross-encoders, generative VLMs, and richer retrieval architectures outside the present benchmark. Finally, our primary clustered intervals describe uncertainty over the fixed benchmark structure rather than arbitrary unseen models or operators; crossed-axis resampling is reported as a broader sensitivity analysis in the supplement. These boundaries motivate asymmetric grids, longer and multilingual text, additional model classes, and recovery methods designed explicitly for simultaneous multimodal degradation.

The broader lesson is not simply that two corruptions hurt more than one. The consequential question is when joint behavior can be reconstructed from the marginal measurements robustness benchmarks already provide. TandemBench makes those quantities explicit and releases the evaluation records, corruption implementations, configurations, and protocol needed to determine where reconstruction succeeds and where direct joint evaluation remains necessary. Corrupted image sets need not be redistributed: deterministic generation scripts and configurations reconstruct the evaluated conditions from the public source data, while the released records preserve the complete benchmark outputs for independent analysis.

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